

Part A: What the tracker reports for one body

$$\begin{aligned} F_{\text{trak}}^* &= F_{\text{trak}} \Delta F_{\text{trak}} \\ &= [R_{\text{trak}}, \vec{p}_{\text{trak}}] [\Delta R_{\text{trak}}, \Delta \vec{p}_{\text{trak}}] \\ &= [R_{\text{trak}} \Delta R_{\text{trak}}, R_{\text{trak}} \Delta \vec{p}_{\text{trak}} + \vec{p}_{\text{trak}}] \end{aligned}$$

$$\begin{aligned} (F_{\text{trak}}^*)^{-1} &= [(R_{\text{trak}}, \Delta R_{\text{trak}})^{-1}, -(R_{\text{trak}}, \Delta R_{\text{trak}})^{-1} (R_{\text{trak}}, \Delta \vec{p}_{\text{trak}} + \vec{p}_{\text{trak}})] \\ &= [\Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1}, -\Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} (R_{\text{trak}}, \Delta \vec{p}_{\text{trak}} + \vec{p}_{\text{trak}})] \\ &= [\Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1}, -\Delta R_{\text{trak}}^{-1} \Delta \vec{p}_{\text{trak}} - \Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} \vec{p}_{\text{trak}}] \end{aligned}$$

$$F_m^* = (F_{\text{trak}}^*)^{-1} F_m \Rightarrow F_m = (F_{\text{trak}}^*)^{-1} F_m^*$$

$$\begin{aligned} F_m &= [\Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1}, -\Delta R_{\text{trak}}^{-1} \Delta \vec{p}_{\text{trak}} - \Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} \vec{p}_{\text{trak}}] [R_m, \vec{p}_m] \\ &= [\Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} R_m, \Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} \vec{p}_m - \Delta R_{\text{trak}}^{-1} \Delta \vec{p}_{\text{trak}} - \Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} \vec{p}_{\text{trak}}] \end{aligned}$$

Part-B: pointer relative to the reference body

$$F_{ref} = [\Delta R_{tank}^{-1} R_{tank}^{-1} R_{ref}, \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{ref} - \Delta R_{tank}^{-1} \Delta \vec{p}_{tank} - \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{tank}]$$

$$F_{pt} = [\Delta R_{tank}^{-1} R_{tank}^{-1} R_{pt}, \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{pt} - \Delta R_{tank}^{-1} \Delta \vec{p}_{tank} - \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{tank}]$$

$$F_{ref}^{-1} = [R_{ref}, p_{ref}]$$

$$R_{ref} = R_{ref}^{-1} R_{tank} \Delta R_{tank}$$

$$p_{ref} = -R_{ref}^{-1} R_{tank} \Delta R_{tank} (\Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{ref} - \Delta R_{tank}^{-1} \Delta \vec{p}_{tank} - \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{tank})$$

$$= -R_{ref}^{-1} R_{tank} \Delta R_{tank} \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{tank} \Delta R_{tank} \Delta R_{tank}^{-1} \Delta \vec{p}_{tank} + R_{ref}^{-1} R_{tank} \Delta R_{tank} \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{tank}$$

$$= -R_{ref}^{-1} R_{tank} R_{tank}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{tank} \Delta \vec{p}_{tank} + R_{ref}^{-1} R_{tank} R_{tank}^{-1} \vec{p}_{tank}$$

$$= -R_{ref}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{tank} + R_{ref}^{-1} \vec{p}_{tank}$$

$$F_{ref}^{-1} = [R_{ref}^{-1} R_{task} \Delta R_{task} - R_{ref}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{task} \Delta \vec{p}_{task} + R_{ref}^{-1} \vec{p}_{task}]$$

$$\begin{aligned} F_{ref}^{-1} F_{ptn} &= [(R_{ref}^{-1} R_{task} \Delta R_{task}) (\Delta R_{task}^{-1} R_{task}^{-1} R_{ptn}) + \\ &\quad R_{ref}^{-1} R_{task} \Delta R_{task} (\Delta R_{task}^{-1} R_{task}^{-1} \vec{p}_{ptn} - \Delta R_{task}^{-1} \vec{p}_{task} \\ &\quad - \Delta R_{task}^{-1} R_{task}^{-1} \vec{p}_{task}) - \\ &\quad R_{ref}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{task} \Delta \vec{p}_{task} + R_{ref}^{-1} \vec{p}_{task}] \\ &= [R_{ref}^{-1} R_{ptn}, R_{ref}^{-1} \vec{p}_{ptn} - R_{ref}^{-1} R_{task} \Delta \vec{p}_{task} - R_{ref}^{-1} \vec{p}_{task} \\ &\quad - R_{ref}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{task} \Delta \vec{p}_{task} + R_{ref}^{-1} \vec{p}_{task}] \\ &= [R_{ref}^{-1} R_{ptn}, R_{ref}^{-1} \vec{p}_{ptn} - R_{ref}^{-1} \vec{p}_{ref}] \end{aligned}$$

Let b be the tip in the tracker coordinates

$$\begin{aligned}
 \vec{v} + \Delta \vec{v} &= F_{\text{trak}} \Delta F_{\text{trak}} \vec{b} \\
 &= [R_{\text{trak}} \Delta R_{\text{trak}}, R_{\text{trak}} \Delta \vec{p}_{\text{trak}} + \vec{p}_{\text{trak}}] b \\
 &= R_{\text{trak}} \Delta R_{\text{trak}} \vec{b} + R_{\text{trak}} \Delta \vec{p}_{\text{trak}} + \vec{p}_{\text{trak}} \\
 &= R_{\text{trak}} (I + \text{sk}(\alpha_{\text{trak}})) \vec{b} + R_{\text{trak}} \vec{G}_{\text{trak}} + \vec{p}_{\text{trak}} \\
 &= R_{\text{trak}} b + R_{\text{skew}}(\alpha_{\text{trak}}) \vec{b} + R_{\text{trak}} \vec{G}_{\text{trak}} + \vec{p}_{\text{trak}} \\
 \Delta v &= R_{\text{trak}} \text{sk}(\alpha_{\text{trak}}) \vec{b} + R_{\text{trak}} \vec{G}_{\text{trak}} \quad \left\{ v = R_{\text{trak}} \vec{b} + \vec{p}_{\text{trak}} \right\} \\
 &= -R_{\text{trak}} \text{sk}(\vec{b}) \alpha_{\text{trak}} + R_{\text{trak}} \vec{G}_{\text{trak}} \\
 &= \begin{bmatrix} -R_{\text{trak}} \text{sk}(\vec{b}) & R_{\text{trak}} \end{bmatrix} \begin{bmatrix} \alpha_{\text{trak}} \\ \vec{G}_{\text{trak}} \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \|\Delta \vec{v}\| &\leq \|\alpha_{\text{trak}}\| \|\vec{b}\| + \|\vec{G}_{\text{trak}}\| = 0.01 \times 1500 + 5 \\
 &= 20 \text{ mm}
 \end{aligned}$$

This is ten times the 2mm specification, relative measurement removes it completely.

The body has k markers $1 \dots k$.

a_k - nominal position of marker

b_k - measured position in tracker coordinates

Consider the unknown in the frame
 $F = [R, p]$

If markers were exactly at nominal positions and tracker had no noise.

$$R \vec{a}_k + \vec{p} = \vec{b}_k$$

Residual of marker

$$\begin{aligned} \vec{e}_k &= F \vec{a}_k - \vec{b}_k \\ &= R \vec{a}_k + \vec{p} - \vec{b}_k \end{aligned}$$

Minimize

$$\sum_{k=1}^k \|\vec{e}_k\|^2 = (R \vec{a}_k + \vec{p} - \vec{b}_k)^T (R \vec{a}_k + \vec{p} - \vec{b}_k)$$

$$\begin{aligned} \{ \vec{u}_k = R \vec{a}_k - \vec{b}_k \} &= (\vec{u}_k + \vec{p})^T (\vec{u}_k + \vec{p}) \\ &= \vec{u}_k^T \vec{u}_k + \vec{u}_k^T \vec{p} + \vec{p}^T \vec{u}_k + \vec{p}^T \vec{p} \\ &= \vec{u}^T \vec{u}_k + 2 \vec{u}^T \vec{p} + \vec{p}^T \vec{p} \end{aligned}$$

differentiate w.r.t $\vec{p}$

$$\frac{\partial (\vec{u}^T \vec{u}_k)}{\partial \vec{p}} = 0$$

$$\frac{\partial}{\partial \vec{p}} (2 \vec{u}_k^T \vec{p}) = 2 \vec{u}_k$$

$$\frac{\partial}{\partial \vec{p}} (\vec{p}^T \vec{p}) = 2 \vec{p}$$

$$\begin{aligned} \frac{\partial \|\vec{e}_k\|^2}{\partial \vec{p}} &= 2 \vec{u}_k + 2 \vec{p} \\ &= 2 (R \vec{a}_k - \vec{b}_k + \vec{p}) \end{aligned}$$

Since there are multiple markers

$$\frac{\partial \|\vec{e}_k\|^2}{\partial \vec{p}} = 2 \sum_{k=1}^K (R \vec{a}_k - \vec{b}_k + \vec{p}) = 0$$

$$\Rightarrow R \sum_{k=1}^K \vec{a}_k - \sum_{k=1}^K \vec{b}_k + K \vec{p} = 0$$

$$\begin{aligned} \vec{p} &= \frac{1}{K} \sum_{k=1}^K \vec{b}_k - R \frac{1}{K} \sum_{k=1}^K \vec{a}_k \\ &= \bar{\vec{b}} - R \bar{\vec{a}} \end{aligned}$$

$$\bar{\vec{a}} = \frac{1}{K} \sum_{k=1}^K \vec{a}_k$$

$$\bar{\vec{b}} = \frac{1}{K} \sum_{k=1}^K \vec{b}_k$$

$$\begin{aligned} \vec{e}_k &= R \vec{a}_k + (\bar{\vec{b}} - R \bar{\vec{a}}) - \vec{b}_k \\ &= R(\vec{a}_k - \bar{\vec{a}}) - (\vec{b}_k - \bar{\vec{b}}) \\ &= R \tilde{\vec{a}}_k - \tilde{\vec{b}}_k \end{aligned}$$

$$\begin{aligned} \tilde{\vec{a}}_k &= \vec{a}_k - \bar{\vec{a}} \\ \tilde{\vec{b}}_k &= \vec{b}_k - \bar{\vec{b}} \end{aligned}$$

$$\begin{aligned}
\|e_k\|^2 &= (R\tilde{a}_k - \tilde{b}_k)^T (R\tilde{a}_k - \tilde{b}_k) \\
&= \tilde{a}_k^T R^T R \tilde{a}_k - \tilde{a}_k^T R^T \tilde{b}_k - \tilde{b}_k^T R \tilde{a}_k + \tilde{b}_k^T \tilde{b}_k \\
&= \tilde{a}_k^T \tilde{a}_k - 2 \tilde{b}_k^T R \tilde{a}_k + \tilde{b}_k^T \tilde{b}_k
\end{aligned}$$

Sum over the markers

$$\rightarrow \sum_{k=1}^K \|\tilde{a}_k\|^2 + \sum_{k=1}^K \|\tilde{b}_k\|^2 - 2 \sum_{k=1}^K \tilde{b}_k^T R \tilde{a}_k$$

$$\text{Maximize } \sum_{k=1}^K \tilde{b}_k^T R \tilde{a}_k$$

$$\tilde{b}_k^T (R\tilde{a}_k) = \text{tr}(\tilde{b}_k^T R\tilde{a}_k) = \text{tr}(R\tilde{a}_k \tilde{b}_k^T)$$

$$\sum_{k=1}^K \tilde{b}_k^T R\tilde{a}_k = \sum_{k=1}^K \text{tr}(R\tilde{a}_k \tilde{b}_k^T)$$

$$= \text{tr}\left(R \sum_{k=1}^K \tilde{a}_k \tilde{b}_k^T\right)$$

$$= \text{tr}(RH)$$

$$\{H = \sum_{k=1}^K \tilde{a}_k \tilde{b}_k^T\}$$

$$= \text{tr}(RUSV^T)$$

$$= \text{tr}(SV^T RU)$$

$$= \text{tr}(SM)$$

$$\begin{cases} H = USV^T \\ M = V^T RU \end{cases}$$

M is orthogonal

$$\begin{aligned} M^T M &= (V^T R U)^T (V^T R U) \\ &= U^T R^T V V^T R U \\ &= U^T R^T R U \\ &= U^T U = I \end{aligned}$$

$$\det(SM) = s_1 M_{11} + s_2 M_{22} + s_3 M_{33} \leq s_1 + s_2 + s_3$$

Since $V^T R U = I$

$$V V^T R U U^T = V I U^T$$

$$R = V U^T$$

The bound is reached only when $M = I$

$V U^T$ is orthogonal, so its determinant is ± 1 or -1 . If its -1 , its mirror image.

Flip the sign of direction with singular value to get proper rotation.

$$R = V \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(V U^T) \end{bmatrix} U^T$$

$$\vec{p} = \vec{b} - \left[v \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(VV^T) \end{bmatrix} v^T \right] \vec{a}$$

So now for every tracker observation we can determine a complete rigid transformation.

$$F = \begin{bmatrix} R & \vec{p} \\ 0 & 1 \end{bmatrix}$$

What marker manufacturing error
does to a fitted frame

The offset is same at every pose

$$\vec{a}_k^* = \vec{a}_B + \vec{\delta}_k$$

$$\vec{b}_k = F_{\text{trk}}^* \vec{a}_k^* = R_{\text{trk}}^* \vec{a}_k^* + \vec{p}_{\text{trk}}^*$$

We use nominal positions and minimise C

$$C = \sum_{k=1}^K \| R \vec{a}_k + \vec{p} - \vec{b}_k \|^2$$

Actual pose

$$\begin{aligned} F_{\text{trk}}^* &= F \Delta F \\ &= [R, \vec{p}] [\Delta R, \Delta \vec{p}] \\ &= [R \Delta R, R \Delta \vec{p} + \vec{p}] \end{aligned}$$

$$\vec{b}_k = R \Delta R \vec{a}_k^* + R \Delta \vec{p} + \vec{p}$$

$$\begin{aligned} R \vec{a}_k + \vec{p} - \vec{b}_k &= R \vec{a}_k + \vec{p} - R \Delta R \vec{a}_k^* - R \Delta \vec{p} - \vec{p} \\ &= -R (\Delta R \vec{a}_k^* + \Delta \vec{p} - \vec{a}_k) \end{aligned}$$

$$C = \sum_{k=1}^K \| \Delta R \vec{a}_k^* + \Delta \vec{p} - \vec{a}_k \|^2$$

ΔF that minimises C depends only on the nominal and actual positions, so it is one fixed frame per body:

$$F_{\text{trk}}^* = F \Delta F \text{ at every pose}$$

Solving for offset

$$\vec{e}_k = \Delta R \vec{a}_k^* + \Delta \vec{p} - \vec{a}_k$$

$$C = \sum_k \|\vec{e}_k\|^2$$

for frame error i.e small

$$\Delta R = I + \text{sk}(\alpha)$$

$$\Delta \vec{p} = \vec{e}$$

$$\vec{a}_k^* = \vec{a}_k + \vec{s}_k$$

$$\begin{aligned} \vec{e}_k &= (I + \text{sk}(\alpha)) (\vec{a}_k + \vec{s}_k) + \vec{e} - \vec{a}_k \\ &= \vec{a}_k + \vec{s}_k + \text{sk}(\alpha) \vec{a}_k + \text{sk}(\alpha) \vec{s}_k + \vec{e} - \vec{a}_k \\ &= \vec{s}_k + \text{sk}(\alpha) \vec{a}_k + \cancel{\text{sk}(\alpha) \vec{s}_k} + \vec{e} \\ &= \vec{s}_k + \text{sk}(\alpha) \vec{a}_k + \vec{e} \end{aligned}$$

we take centroid of the nominal marker

$$\bar{a} = \frac{1}{K} \sum_{k=1}^K \vec{a}_k$$

$$\tilde{a}_k = \vec{a}_k - \bar{a}$$

$$\vec{a}_k = \tilde{a}_k + \bar{a}$$

$$\begin{aligned} \vec{G}_k &= \vec{S}_k + \text{sk}(\alpha)(\bar{a} + \tilde{a}_k) + \vec{G} \\ &= \vec{S}_k + \text{skew}(\alpha) \bar{a} + \text{skew}(\alpha) \tilde{a}_k + \vec{G} \\ &= \vec{S}_k + \text{skew}(\alpha) \tilde{a}_k + \vec{G}_c \end{aligned}$$
$$\begin{aligned} \{\vec{G}_c &= \vec{G} + \text{sk}(\alpha) \bar{a}\} \\ &= \begin{bmatrix} -\text{skew}(\tilde{a}_k) & \mathbf{I} \end{bmatrix} \begin{bmatrix} \alpha \\ \vec{G}_c \end{bmatrix} + \vec{S}_k \end{aligned}$$

Centroid positions

$$\begin{aligned} \tilde{a}_k &= \vec{a}_k - \bar{a} \\ \sum_{k=1}^K \tilde{a}_k &= \sum_{k=1}^K (\vec{a}_k - \bar{a}) \\ &= \sum_{k=1}^K \vec{a}_k - \sum_{k=1}^K \bar{a} \end{aligned}$$

$$\sum_k^k \bar{a} = K \bar{a}$$

$$\bar{a} = \frac{1}{K} \sum_{k=1}^k \vec{a}_k$$

$$\sum \vec{a}_k = K \bar{a}$$

$$\sum_{k=1}^k \tilde{a}_k = K \bar{a} - K \bar{a} = 0$$

$$\vec{E}_k = \vec{S}_k - sk(\tilde{a}_k) \alpha + \vec{E}_c$$

differentiate w.r.t $\vec{E}_c$
since

$$\frac{\partial \vec{E}_k}{\partial \vec{E}_c} = I$$

we have to minimize

$$\frac{\partial E}{\partial \vec{E}_c} = 2 \sum_{k=1}^k \vec{E}_k \quad \left\{ E = \sum_{k=1}^k \|\vec{E}_k\|^2 \right\}$$

$$\sum_{k=1}^k (\vec{S}_k - sk(\tilde{a}_k) \alpha + \vec{E}_c) = 0$$

$$\sum sk(\tilde{a}_k) = sk\left(\sum_k \tilde{a}_k\right) = sk(0) = 0$$

$$\sum_k \vec{E}_c = K \vec{E}_c$$

$$\sum_k \vec{\delta}_k + K \vec{E}_c = 0$$

$$K \vec{E}_c = - \sum_k \vec{\delta}_k$$

$$\vec{E}_c = - \frac{1}{K} \sum_{k=1}^K \vec{\delta}_k$$

frames translation at marker centroid is simply the negative of the avg displacement

If all markers have shifted in one direction then body shifts in the opposite direction to compensate.

Rotational offset

$$\vec{w}_k = \vec{\delta}_k + \vec{E}_c$$

$$\vec{E}_k = \vec{w}_k - \text{skew}(\tilde{\alpha}_k) \alpha$$

$$\begin{aligned} \|\vec{E}_k\|^2 &= (\vec{w}_k - S_k \alpha)^T (\vec{w}_k - S_k \alpha) \\ &= \vec{w}_k^T \vec{w}_k - 2 \vec{w}_k^T S_k \alpha + \alpha^T S_k^T S_k \alpha \end{aligned} \quad \left\{ \begin{array}{l} \text{skew}(\tilde{\alpha}_k) \\ = S_k \end{array} \right\}$$

differentiate w.r.t α

$$\frac{\partial \|\epsilon_k\|^2}{\partial \alpha} = -2 S_k^T \vec{w}_k + 2 S_k^T S_k \alpha$$

$$\text{Since } S_k^T = -S_k$$

$$S_k^T S_k \alpha = -S_k^2 \alpha$$

$$S_k \vec{e}_k = S_k \vec{w}_k - S_k^2 \alpha$$

$$\frac{\partial \|\vec{e}_k\|^2}{\partial \vec{\alpha}} = 2 S_k \vec{e}_k$$

$$E = \sum_k \|\vec{e}_k\|^2$$

$$\frac{\partial E}{\partial \alpha} = 2 \sum_{k=1}^K S_k \vec{e}_k = 0$$

$$\sum_k S_k (\vec{S}_k - S_k \alpha + \vec{e}_0) = 0$$

$$\sum_k S_k \vec{S}_k - \sum_k S_k^2 \alpha + \sum_k S_k \vec{e}_c = 0$$

$$\sum_k S_k \vec{e}_c = \left(\sum_k S_k \right) \vec{e}_c$$

$$\sum_k S_k = S_k \left(\sum_k \tilde{a}_k \right) = 0$$

$$\sum_k S_k \vec{S}_k - \left(\sum_k S_k^2 \right) \alpha = 0$$

$$\left(\sum_k S_k^2 \right) \alpha = \sum_k S_k \vec{S}_k = \sum_k \text{skew}(\tilde{a}_k) \vec{S}_k$$

$$\alpha = \left[\sum_{k=1}^K \text{sk}(\tilde{a}_k)^2 \right]^{-1} \sum_{k=1}^K \text{sk}(\tilde{a}_k) \vec{S}_k$$

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$\text{sk}(\vec{a}) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$

$$\text{sk}(\vec{a})^2 = \begin{bmatrix} -a_2^2 - a_3^2 & a_1 a_2 & a_1 a_3 \\ a_1 a_2 & -a_1^2 - a_3^2 & a_2 a_3 \\ a_1 a_3 & a_2 a_3 & -a_1^2 - a_2^2 \end{bmatrix}$$

$$\vec{a} \vec{a}^T = \begin{bmatrix} a_1^2 & a_1 a_2 & a_1 a_3 \\ a_1 a_2 & a_2^2 & a_2 a_3 \\ a_1 a_3 & a_2 a_3 & a_3^2 \end{bmatrix}$$

$$\|\vec{a}\|^2 \mathbf{I} = a_1^2 + a_2^2 + a_3^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{sk}(\vec{a})^2 = \vec{a} \vec{a}^T - \|\vec{a}\|^2 \mathbf{I}$$

$$-\text{sk}(\vec{a})^2 = \|\vec{a}\|^2 \mathbf{I} - \vec{a} \vec{a}^T$$

$$\begin{aligned} \alpha &= - \left[\sum_{k=1}^K (\|\vec{a}_k\|^2 \mathbf{I} - \vec{a}_k \vec{a}_k^T) \right]^{-1} \sum_{k=1}^K \vec{a}_k \times \vec{S}_k \\ &= - \mathbf{J}^{-1} \sum_k \text{sk}(\tilde{a}_k) \vec{S}_k \end{aligned}$$

$$\begin{aligned}
\text{Cov}(\alpha) &= \left[-\mathcal{J}^{-1} \sum_k \text{sk}(\tilde{a}_k) \right] \sigma_a^2 \mathbf{I} \left[-\mathcal{J}^{-1} \sum_k \text{sk}(\tilde{a}_k) \right]^T \\
&= \sigma_a^2 \mathcal{J}^{-1} \sum_k \text{sk}(\tilde{a}_k) \text{sk}(\tilde{a}_k)^T \mathcal{J}^{-1} \\
&= \sigma_a^2 \mathcal{J}^{-1} \left[-\sum_k \text{sk}(\tilde{a}_k)^2 \right] \mathcal{J}^{-1} \\
&= \sigma_a^2 \mathcal{J}^{-1} \mathcal{J} \mathcal{J}^{-1}
\end{aligned}$$

$$\text{Cov}(\alpha) = \sigma_a^2 \mathcal{J}^{-1}$$

$$\text{Cov}(\vec{\epsilon}_c) = \frac{\sigma_a^2}{K} \mathbf{I}$$

$$\begin{aligned}
\text{Cov}(\alpha, \vec{\epsilon}_c) &= \sum_{k=1}^K \left(-\mathcal{J}^{-1} \text{sk}(\tilde{a}_k) \right) (\sigma_a^2 \mathbf{I}) \left(-\frac{1}{K} \mathbf{I} \right)^T \\
&= \frac{\sigma_a^2}{K} \mathcal{J}^{-1} \sum_{k=1}^K \text{sk}(\tilde{a}_k) = 0
\end{aligned}$$

Marker Refinement

For reading n

Actual body pose relative to torso

$$F_{\text{trk},n}^* = [R_{\text{trk},n}^*, \vec{P}_{\text{trk},n}^*]$$

$$F_n = [R_n, \vec{P}_n]$$

$$F_{\text{trk},n}^* = F_n \Delta F$$

$$\Delta F = [\Delta R, \Delta \vec{P}]$$

Actual marker position a_k^*

$$\vec{b}_{k,n} = R_{\text{trk},n}^* \vec{a}_k^* + \vec{P}_{\text{trk},n}^* + \vec{v}_{k,n}$$

$$= R_n \Delta R \vec{a}_k^* + R_n \Delta \vec{P} + \vec{P}_n + \vec{v}_{k,n} \quad \left\{ \begin{array}{l} \vec{v}_{k,n} = \text{noise?} \\ + \vec{v}_{k,n} \end{array} \right.$$

$$F_n^{-1} \vec{b}_{k,n} = R_n^{-1} (\vec{b}_{k,n} - \vec{P}_n)$$

$$= R_n^{-1} (R_n \Delta R \vec{a}_k^* + R_n \Delta \vec{P} + \vec{P}_n + \vec{v}_{k,n} - \vec{P}_n)$$

$$= R_n^{-1} R_n \Delta R \vec{a}_k^* + R_n^{-1} R_n \Delta \vec{P} + R_n^{-1} \vec{v}_{k,n}$$

$$= \Delta R \vec{a}_k^* + \Delta \vec{P} + R_n^{-1} \vec{v}_{k,n}$$

The body was in a different pose for every reading: $R_1, R_2, \dots$

$$\Rightarrow \Delta R \vec{a}_k^* + \Delta \vec{p}$$

$$\text{Cov}(\vec{v}_{k,n}) = R_n^{-1} \text{Cov}(\vec{v}_{k,n}) (R_n^{-1})^T$$

$$\begin{aligned} \text{Cov}(\vec{v}_{k,n}) &= \sigma_b^2 I \\ &= R_n^{-1} (\sigma_b^2 I) (R_n^{-1})^T \\ &= \sigma_b^2 R_n^{-1} (R_n^{-1})^T \end{aligned}$$

$$\text{Cov}(R_n^{-1} \vec{v}_{k,n}) = \sigma_b^2 I$$

For every reading

$$F_n^{-1} \vec{b}_{k,n} = \Delta R \vec{a}_k^* + \Delta \vec{p} + R_n^{-1} \vec{v}_{k,n}$$

refined estimate

$$\begin{aligned} \vec{a}_k &= \frac{1}{N} \sum_{n=1}^N F_n^{-1} \vec{b}_{k,n} \\ &= \frac{1}{N} \sum_{n=1}^N (\Delta R \vec{a}_k^* + \Delta \vec{p} + R_n^{-1} \vec{v}_{k,n}) \\ &= \Delta R \vec{a}_k^* + \Delta \vec{p} + \frac{1}{N} \sum_{n=1}^N R_n^{-1} \vec{v}_{k,n} \end{aligned}$$

$$\begin{aligned}\text{Cov}(\hat{\vec{a}}_k) &= \sum_{n=1}^N \frac{1}{N} \mathbf{I} \left(\sigma_b^2 \mathbf{I} \right) \left(\frac{1}{N} \mathbf{I} \right)^T \\ &= \frac{\sigma_b^2}{N} \mathbf{I}\end{aligned}$$

Shape improves?

$$\hat{\vec{a}}_j = \Delta R \vec{a}_j^* + \Delta \vec{p}$$

$$\hat{\vec{a}}_k = \Delta R \vec{a}_k^* + \Delta \vec{p}$$

$$\begin{aligned}\hat{\vec{a}}_j - \hat{\vec{a}}_k &= \Delta R \vec{a}_j^* - \Delta R \vec{a}_k^* \\ &= \Delta R (\vec{a}_j^* - \vec{a}_k^*)\end{aligned}$$

$$\|\hat{\vec{a}}_j - \hat{\vec{a}}_k\| = \|\Delta R (\vec{a}_j^* - \vec{a}_k^*)\|$$

Relative geometry between
markers is recovered

$$\mathbf{F}' = [\mathbf{R}', \vec{p}']$$

fitting residual

$$R \hat{\vec{a}}_k + \vec{p}' - \vec{b}_k$$

Ignore tiny errors

$$\hat{a}_k = \Delta R \vec{a}_k^* + \Delta \vec{p}$$

Then

$$R'(\Delta R \vec{a}_k^* + \Delta \vec{p}) + \vec{p}' = (R_{\text{trk}}^* \vec{a}_k^* + \vec{p}_{\text{trk}}^*)$$

$$= R' \Delta R \vec{a}_k^* + R' \Delta \vec{p} + \vec{p} - R_{\text{trk}}^* \vec{a}_k^* - \vec{p}_{\text{trk}}^*$$

$$R' \hat{a}_k + \vec{p}' - \vec{b}_k = (R' \Delta R - R_{\text{trk}}^*) \vec{a}_k^* + (R' \Delta \vec{p} + \vec{p}' - \vec{p}_{\text{trk}}^*)$$

For residual to be zero, we need

$$R' \Delta R = R_{\text{trk}}^*$$

$$R' \Delta \vec{p} + \vec{p}' = \vec{p}_{\text{trk}}^*$$

These give

$$F' \Delta F = F_{\text{trk}}^*, \quad F' = F$$

After refinement, the frame hasn't

suddenly become true physical frame,

the ΔF is in there.

The pointers offset does no harm

$$\vec{w} = R_{\text{trk, ptr}}^* \vec{p}_{\text{tip}}^* + \vec{p}_{\text{trk, ptr}}^*$$

$$\left. \begin{aligned} \Delta R_{\text{ptr}} \vec{p}_{\text{ptr}}^* + \Delta \vec{p}_{\text{ptr}} \end{aligned} \right\} = R_{\text{ptr}} \Delta R_{\text{ptr}} \vec{p}_{\text{tip}}^* + R_{\text{ptr}} \Delta \vec{p}_{\text{ptr}} + \vec{p}_{\text{ptr}}$$

$$= R_{\text{ptr}} \vec{p}_{\text{tip}} + \vec{p}_{\text{ptr}} = F_{\text{ptr}} \vec{p}_{\text{tip}}$$

$\vec{p}_{tip}$ is constant: the tip expressed in the pointer's frame. The calibration estimates it, the test uses it, so ΔF_{Pto} never appears.

Part C: Finding pointer tip

$$F_{rel} = F_{rel}^{-1} F_{ptn}$$

$$\vec{p}_{tip}^{rel} = R_{rel} \vec{p}_{tip} + p_{rel}$$

Unknown point fixed to the pointer - $\vec{p}_{tip}$

$$\vec{b}_{tip} = R_{ptn} \vec{p}_{tip} + \vec{p}_{ptn}$$

$$\left\{ \begin{array}{l} \text{same as} \\ b_k = R \vec{a}_k + \vec{p} \end{array} \right\}$$

When surgeon pivots the pointer around the calibration sensor, the tip remains at one physical location

fixed point - $\vec{p}_{pivot}$

$$R_i \vec{p}_{tip} + \vec{p}_i = \vec{p}_{pivot} \quad (\text{observation } i)$$

$$R_j \vec{p}_{tip} + \vec{p}_j = \vec{p}_{pivot}$$

Group terms

$$(R_i - R_j) \vec{p}_{tip} + (\vec{p}_i - \vec{p}_j) = 0$$

$$(R_i - R_j) \vec{p}_{tip} = \vec{p}_j - \vec{p}_i \quad (\text{calibration equation})$$

$$\vec{P}_{tip} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

what does one observation do?

$$(R_i - R_1) \vec{P}_{tip} = \vec{P}_1 - \vec{P}_i$$

$$(R_i - R_1) \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = \begin{bmatrix} p_{1x} - p_{ix} \\ p_{1y} - p_{iy} \\ p_{1z} - p_{iz} \end{bmatrix}$$

$$a_{11} p_x + a_{12} p_y + a_{13} p_z = p_{1x} - p_{ix}$$

$$a_{21} p_x + a_{22} p_y + a_{23} p_z = p_{1y} - p_{iy}$$

$$a_{31} p_x + a_{32} p_y + a_{33} p_z = p_{1z} - p_{iz}$$

One pair of pointer observations gives us three equations involving three unknown components of tip

But this isn't enough. It's not about the number of equations but about the independent information the equations contain.

Eg. Let's say the pointer barely changes orientation, $R_i \approx R_1$

$$R_i - R_1 \approx 0$$

our equation becomes

$$0 \cdot \vec{p}_{tip} \approx \vec{p}_1 - \vec{p}_i$$

That leaves us with no information about $\vec{p}_{tip}$

We want the rotational diversity between observations

Different pointer observations provide the information needed to determine $\vec{p}_{tip}$

So we add more observations
Using observation 1 as our reference

$$(R_2 - R_1) \vec{p}_{tip} = \vec{p}_1 - \vec{p}_2$$

$$(R_N - R_1) \vec{p}_{tip} = \vec{p}_1 - \vec{p}_N$$

we now have many equations for
the same unknown vector

Stack the

$$M_i = R_i - R_1$$

$$\vec{b}_i = \vec{p}_1 - \vec{p}_i$$

$$M_i \vec{p}_{tip} = \vec{b}_i$$

$$M_i = \begin{bmatrix} R_2 - R_1 \\ \vdots \\ R_N - R_1 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} \vec{p}_1 - \vec{p}_2 \\ \vdots \\ \vec{p}_1 - \vec{p}_N \end{bmatrix}$$

$$M \in \mathbb{R}^{3(N-1) \times 3}$$

$$b \in \mathbb{R}^{3(N-1)}$$

$$\mathbb{R}^{87 \times 3}$$

{ for 30 observations }

Minimizing

$$\vec{e} = M \vec{p}_{tip} - \vec{b}$$

$$J(\vec{p}_{tip}) = \| M \vec{p}_{tip} - \vec{b} \|^2$$

$$\hat{\vec{p}}_{tip} = \arg \min_{\vec{p}_{tip}} \| M \vec{p}_{tip} - \vec{b} \|^2$$

$$\begin{aligned} J(\vec{p}_{tip}) &= (M \vec{p}_{tip} - \vec{b})^T (M \vec{p}_{tip} - \vec{b}) \\ &= (M \vec{p}_{tip})^T (M \vec{p}_{tip}) - (M \vec{p}_{tip})^T \vec{b} - \vec{b}^T (M \vec{p}_{tip}) + \vec{b}^T \vec{b} \\ &= \vec{p}_{tip}^T M^T M \vec{p}_{tip} - 2 \vec{p}_{tip}^T M^T \vec{b} + \vec{b}^T \vec{b} \end{aligned}$$

$$\frac{\partial J}{\partial \vec{p}_{tip}} = 0$$

$$2 M^T M \vec{p}_{tip} = 2 M^T \vec{b}$$

$$\hat{\vec{p}}_{\text{tip}} = (\mathbf{M}^T \mathbf{M})^{-1} \mathbf{M}^T \mathbf{b}$$

Ideal pivot equation

$$t_{\text{eff},i} = \vec{p}_{\text{sensed},i} - \vec{p}_i$$

$$\vec{p}_{\text{sensed},i} = R_i \vec{p}_{\text{tip}} + \vec{p}_i$$

$$t_{\text{eff},i} = \vec{p}_{\text{sensed},i} - \vec{p}_i = R_i \vec{p}_{\text{tip}}$$

$$R_i \vec{p}_{\text{tip}} - R_1 \vec{p}_{\text{tip}} = t_{\text{eff},i} - t_{\text{eff},1}$$

$$(R_i - R_1) \vec{p}_{\text{tip}} = t_{\text{eff},i} - t_{\text{eff},1}$$

We now know that we at least need 2 observations and our experiment uses 30 observations and provides many more equations than unknowns, while different orientations make the system overdetermined.

Suppose the ideal relationship is

$$M \vec{p}_{\text{tip}} = b_{\text{true}}$$

$$M \vec{p}_{\text{tip}} = b - \epsilon$$

$$\begin{cases} b = b_{\text{true}} + \epsilon \\ b_{\text{true}} = b - \epsilon \end{cases}$$

Least square estimate

$$\hat{\vec{p}}_{\text{tip}} = (M^T M)^{-1} M^T b$$

$$= (M^T M)^{-1} M^T (M \vec{p}_{\text{tip}} + \epsilon)$$

$$= (M^T M)^{-1} M^T M \vec{p}_{\text{tip}} + (M^T M)^{-1} M^T \epsilon$$

$$\hat{\vec{p}}_{\text{tip}} = \vec{p}_{\text{tip}} + (M^T M)^{-1} M^T \epsilon$$

$$\hat{\vec{p}}_{\text{tip}} - \vec{p}_{\text{tip}} = (M^T M)^{-1} M^T \epsilon$$

Let's take on assume $\text{Cov}(\epsilon) = \sigma^2 I$

$$\hat{\vec{p}}_{\text{tip}} - \vec{p}_{\text{tip}} = (M)^+ \epsilon$$

$$\text{Cov}((M)^+ \epsilon) = M^+ \text{Cov}(\epsilon) (M^+)^T$$

$$= M^+ (\sigma^2 I) (M^+)^T$$

$$= \sigma^2 (M^+ (M^+)^T)$$

$$= \sigma^2 ((M^T M)^{-1} M^T M (M^T M)^{-1})$$

$$= \sigma^2 (M^T M)^{-1}$$

$$\text{Cov}(\hat{p}_{\text{tip}}) = \sigma^2 (M^T M)^{-1}$$

from residual

$$e = M \hat{p}_{\text{tip}} - b$$

there are $m = 3(N-1)$ scalar equations
in our stacked system.

we estimated three unknown quantities
 $\text{dof} = m - 3$

$$\sigma^2 = \frac{e^T e}{m-3}$$

$$\begin{aligned} \text{Cov}(\hat{p}_{\text{tip}}) &= \frac{e^T e}{m-3} (M^T M)^{-1} \\ &= \frac{e^T e}{87} (M^T M)^{-1} \end{aligned}$$

$$\Sigma = \text{Cov}(\hat{p}_{\text{tip}})$$

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{13} \\ \Sigma_{21} & \Sigma_{22} & \Sigma_{23} \\ \Sigma_{31} & \Sigma_{32} & \Sigma_{33} \end{bmatrix}$$

$$\sum_n = \sigma_x^2$$

$$\sum_{22} = \sigma_y^2$$

$$\sum_{33} = \sigma_z^2$$

So the predicted uncertainty of the calibrated
tip can be $(\sigma_x, \sigma_y, \sigma_z)_{mm}$

Define test error

$$F_{\text{test}} = [R_{\text{test}}, P_{\text{test}}]$$

estimated tip position

$$P_{\text{est}} = R_{\text{test}} \hat{P}_{\text{tip}} + P_{\text{test}}$$

simulation provides the ground truth tip
position

$$P_{\text{true}}$$

Error

$$e_{\text{test}} = \|P_{\text{test}} - P_{\text{true}}\|$$

If we have N test poses, then

$$e_{\text{mean}} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} e_{\text{test}, i}$$

maximum error

$$e_{\text{max}} = \max_i e_{\text{test}, i}$$

$$e_{\text{test}} \leq 2 \text{ mm}$$

$$\vec{p}_{\text{est}} = F_{\text{ref}}^{-1} \vec{x} = R_{\text{ref}}^{-1} (\vec{x} - \vec{p}_{\text{ref}})$$

$$\begin{aligned} \vec{p}_{\text{true}} &= (F_{\text{true, ref}}^*)^{-1} \vec{x} = (R_{\text{true, ref}}^*)^{-1} (\vec{x} - \vec{p}_{\text{true, ref}}^*) \\ &= (R_{\text{ref}} \Delta R_{\text{ref}})^{-1} (\vec{x} - R_{\text{ref}} \Delta \vec{p}_{\text{ref}} - \vec{p}_{\text{ref}}) \\ &= \Delta R_{\text{ref}}^{-1} (R_{\text{ref}}^{-1} (\vec{x} - \vec{p}_{\text{ref}}) - R_{\text{ref}}^{-1} R_{\text{ref}} \Delta \vec{p}_{\text{ref}}) \\ &= \Delta R_{\text{ref}}^{-1} (\vec{p}_{\text{est}} - \Delta \vec{p}_{\text{ref}}) \\ &= (I - \text{skew}(\alpha_{\text{ref}})) (\vec{p}_{\text{est}} - \Delta \vec{p}_{\text{ref}}) \\ &= \vec{p}_{\text{est}} - \vec{e}_{\text{ref}} - \text{skew}(\alpha_{\text{ref}}) \vec{p}_{\text{est}} + \text{skew}(\alpha_{\text{ref}}) \vec{e}_{\text{ref}} \end{aligned}$$

$$\Delta \vec{p}_{\text{est}} = \vec{p}_{\text{true}} - \vec{p}_{\text{est}} = -\alpha_{\text{ref}} \times \vec{p}_{\text{est}} - \vec{e}_{\text{ref}}$$

measure tip from centroid $\vec{a}$ of the reference frame. $\vec{p}_{est} = \vec{a} + \vec{d}$

$$\begin{aligned}\Delta \vec{p}_{est} &= -\alpha_{ref} \times (\vec{a} + \vec{d}) - \vec{e}_{ref} \\ &= -(\alpha_{ref} \times \vec{d} + \vec{e}_{c,ref}) \\ &\quad \left\{ \vec{e}_{c,ref} = \vec{e}_{ref} + \alpha_{ref} \times \vec{a} \right\} \\ &= \begin{bmatrix} -\text{skew}(\vec{d}) & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \alpha_{ref} \\ \vec{e}_{c,ref} \end{bmatrix}\end{aligned}$$

$$= B \vec{\eta}_{ref}$$

from manufacturing error

$$C_{\eta} = \begin{bmatrix} \sigma_a^2 J^{-1} & 0 \\ 0 & \frac{\sigma_a^2}{k} \mathbf{I} \end{bmatrix}$$

$$\begin{aligned}\text{Cov}(\Delta \vec{p}_{est}) &= B C_{\eta} B^T = [\text{sk}(\vec{d}) \quad -\mathbf{I}] C_{\eta} [\text{sk}(\vec{d})^T \quad -\mathbf{I}^T] \\ &= \sigma_a^2 \left(\frac{1}{k} \mathbf{I} + \text{sk}(\vec{d}) J^{-1} \text{sk}(\vec{d})^T \right)\end{aligned}$$